EFY TESTED PREETI BHAKTA

Android Smartphone-Based FUNCTION GENERATOR

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Android smartphones are rapidly becoming popular in the education industry. Some can even be used as signal or function generators. Bulky signal generators available in the market are costly and complex to operate. Android smartphones can be used as signal generators to perform basic tasks. Android-based function generators are not as advanced as standalone function generators, but these are a good option for basic applications.

This Android-based function generator output is in the range of audio frequency signals (20Hz-20,000Hz), available at the audio output port of cellphones. Most cellphones have 3.5mm audio jack output ports. You can use an audio cable with the audio jack between the cellphone and external devices or any other applications.

Signals can also be amplified using a suitable amplifier, if required, for interfacing with other devices. While interfacing with external devices, a proper protection circuit must be used to prevent any damage to the smartphone.

Android-based function generator

Android smartphones are capable of generating audio frequency range signals and tones. Digital signal processing (DSP) hardware available on Android smartphones can be used as a function generator, but with some limitations.

Functions available on the smartphone-based function generator are square-wave, sinewave, triangular-wave, sawtooth-wave and supported frequency range from 20Hz to 20,000Hz. Signals generated using an Android smartphone can be tested using a digital storage oscilloscope (DSO) and Windows PC-based CRO software (Zelscope).

Requirements for function generation

1. Android smartphone
2. Internet access to download apps from Play Store
3. One of the following apps: Function Generator, Frequency Generator, Tone Generator, Sound Generator and Impulse—all available for free download on Play Store

Fig. 1: Function Generator app installed on Android smartphone along with crocodile cables

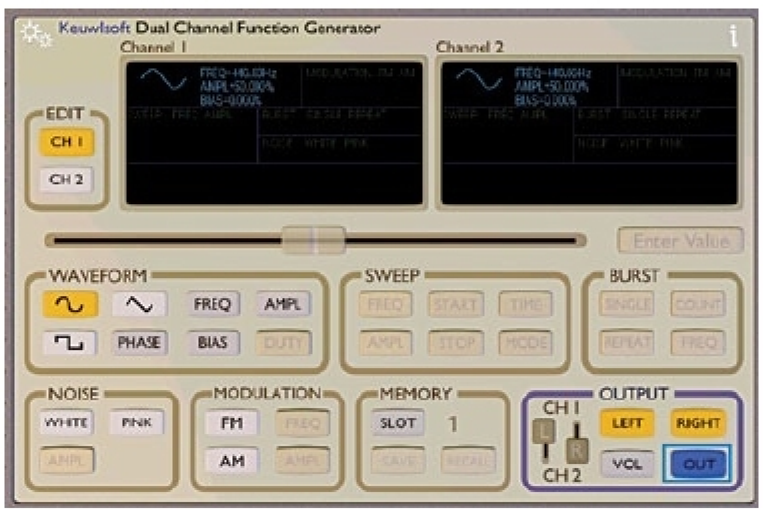

Fig. 2: Function Generator app

Fig. 3: Select channel and waveform

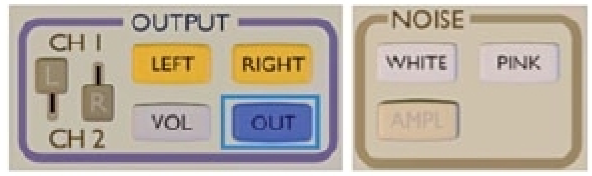

Fig. 4: Select OUT for output signal

Fig. 5: FM and AM modulation

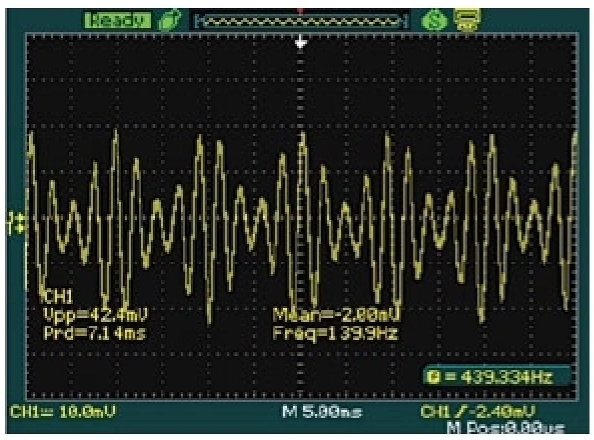

Fig. 6: AM signal of amplitude modulated sinewave as observed on oscilloscope

Fig. 7: 522Hz square-wave signal using Sound Generator app